

Campus Placement Data Cleaning and Exploratory Data Analysis Using Pandas

Project Overview

This project focuses on cleaning and analyzing a campus placement dataset using Python and Pandas. The objective was to identify factors influencing student placements and salary outcomes through data cleaning, missing value treatment, outlier handling, and exploratory data analysis.

Data Cleaning Process

Missing Value Handling

- Identified missing values in the dataset.
- Checked distribution and skewness of numerical columns.
- Used median imputation for skewed distributions.
- Handled missing salary values appropriately.

Duplicate Records

- Checked for duplicate entries and removed them if present.

Outlier Treatment

- Detected outliers using the IQR method.
 - Treated outliers using clipping based on lower and upper bounds.
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Exploratory Data Analysis

Overall Placement Rate

Status	Percentage
Placed	68.84%
Not Placed	31.16%

Insight: More than two-thirds of students were successfully placed, indicating a strong placement performance.

Gender-wise Placement Analysis

Gender	Not Placed	Placed
Female	28	48
Male	39	100

Insight: Male students had a higher number of placements compared to female students in this dataset.

Salary Analysis

- Average Salary: ₹269,778.72
- Highest Salary: ₹331,250.00

Insight: The salary distribution indicates moderate variation among placed students, with the highest package reaching ₹331,250.

Specialization-wise Placement Analysis

Specialization	Not Placed	Placed
Mkt&Fin	25	95
Mkt&HR	42	53

Insight: Students specializing in Marketing & Finance achieved significantly higher placement counts than Marketing & HR students.

Specialization-wise Salary Analysis

Specialization	Average Salary
Mkt&Fin	₹271,664.58
Mkt&HR	₹264,026.32

Insight: Marketing & Finance students earned slightly higher average salaries than Marketing & HR students.

Work Experience Analysis

Work Experience	Not Placed	Placed
No	57	84
Yes	10	64

Insight: Students with prior work experience showed a significantly higher placement success rate, suggesting that experience positively influences employability.

Degree Stream Analysis

Degree Stream	Not Placed	Placed
Comm&Mgmt	43	102
Others	6	5
Sci&Tech	18	41

Insight: Commerce & Management students recorded the highest placement count among all degree streams.

Key Findings

1. Overall placement rate was 68.84%.
 2. Students with prior work experience had significantly better placement outcomes.
 3. Marketing & Finance specialization showed both higher placement counts and higher average salaries.
 4. Commerce & Management students achieved the highest number of placements.
 5. The average salary of placed students was ₹269,778.72.
 6. The highest salary offered was ₹331,250.
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Conclusion

The analysis indicates that work experience, specialization, and degree stream play important roles in placement outcomes. Marketing & Finance students demonstrated stronger placement performance and salary outcomes compared to Marketing & HR students. Additionally, students with prior work experience had a clear advantage during placements.